

Capstone Project

CST8812

GROUP 8 – FINAL REPORT

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Table of Contents

Organization.....	2
Overview.....	2
Objective.....	2
Business Case.....	2
Risks.....	2
Out of Scope	3
Technical Design Document	3
Conceptual Design	6
Storing patient’s medical records in Azure Portal.....	8
Technical Specifications	11
Assumptions.....	14
Issues & Concerns	14
Projected Costs	16
Document Tracking.....	17
Conclusion	17
Appendix A	18
Technical Runbook Document for Patient Records System for Healthcare on MS Azure	18
Appendix B	20
Technical As-Built Document for Patient Records System for Healthcare on MS Azure	20
Appendix C	22
Management of Risks	22
Stakeholders.....	23
Initial Scope	24
Initial Planning.....	24
Work Breakdown Structure.....	25
Acceptance Test	25
Lessons Learned.....	27
Training Plan.....	27

Organization			
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Project

Overview

The project entails the development of an interface that utilizes Microsoft Azure's storage container to securely archive patient medical records in compliance with healthcare regulations like HIPAA. This interface will serve as a centralized repository for storing various medical documents, including test results, medical imaging files, and treatment histories. By leveraging Azure's robust storage capabilities, the aim is to create a scalable and efficient solution for healthcare providers to securely manage and access sensitive patient information.

Objective

The primary objective of the project is to establish a secure and compliant system for archiving patient medical records using Microsoft Azure's storage container. This involves designing and implementing an intuitive interface that allows healthcare professionals to easily upload, retrieve, and manage medical documents while ensuring adherence to HIPAA regulations and maintaining data integrity and confidentiality.

Business Case

The business case for the project centers on the necessity for healthcare organizations to securely manage and store large volumes of patient medical records while complying with stringent regulatory requirements such as HIPAA. By leveraging Azure's storage container, the project aims to offer a cost-effective and scalable solution that minimizes infrastructure overheads and mitigates the risk of data breaches, thereby enhancing trust and confidence among patients and healthcare stakeholders.

Risks

One of the primary risks associated with the project is ensuring the security and privacy of patient information stored within the Azure environment. This involves implementing robust access controls, encryption mechanisms, and regular security audits to safeguard against unauthorized access and data breaches. Additionally, challenges may arise concerning data migration, integration with existing healthcare systems, and ensuring compatibility with various file formats and medical imaging standards.

Out of Scope

While the primary focus is on securely archiving patient medical records using Azure's storage container, certain activities fall outside the scope of the project. This includes the development of specific healthcare applications or electronic medical record (EMR) systems, typically handled by separate initiatives. Additionally, any regulatory compliance measures beyond HIPAA requirements, such as those specific to other regions or jurisdictions, are considered out of scope for the project.

Technical Design Document

1. Azure Service Configuration

Azure Storage Account:

- Create an Azure Storage Account:
 - Subscription: Choose your healthcare organization's subscription.
 - Resource group: Use an existing or create a new one for organization.
 - Storage account name: Follow Azure's naming conventions and ensure uniqueness.
 - Region: Opt for the region offering the best compliance with healthcare regulations like HIPAA.
 - Performance: "Standard" to balance cost and performance.
 - Redundancy: "Geo-redundant storage (GRS)" for disaster recovery compliance.
- Best Practices:
 - Enable "Secure transfer required" for secure data transfer.
 - Utilize "Hot" access tier for frequently accessed patient records and "Cool" or "Archive" for less active data to save costs.

Blob Containers for Patient Records:

- Creating Containers:
 - Use descriptive names like "patient-info", "test-results", "imaging-files", etc.
 - Set to "Private" to maintain confidentiality and compliance.
- Security and Access:
 - Use Azure Active Directory (AD) for authentication.
 - Define Azure role-based access control (RBAC) for tight access management, granting specific permissions to healthcare staff based on their roles.

2. Application Services Layer

- Implement Azure App Services for hosting the web applications that interact with the patient records.
- Use Azure Functions for serverless computing tasks, such as processing data and integrating services.

3. Data Layer

Azure SQL Database:

- Host patient data such as personal info and test results, ensuring all data is encrypted at rest and in transit.

Azure Blob Storage:

- Store large binary files like medical imaging in blob storage.
- Apply data lifecycle management to manage blob retention and archiving.

4. Security Implementation

- Identity and Access Management:
 - Integrate with Azure AD for secure access control.
 - Set granular permissions using custom roles if necessary.
- Network Security:
 - Implement Azure Firewall and Network Security Groups (NSGs) to control inbound and outbound traffic.
 - Use Virtual Network (VNet) service endpoints for secure Azure service connections.
- Data Security:
 - Implement Azure Information Protection for data classification and Azure Key Vault for managing encryption keys.
 - Use Azure Advanced Threat Protection for continuous monitoring and threat detection.

5. Compliance and Monitoring

- Compliance Tools:
 - Utilize Azure Policy and Azure Blueprints to ensure compliance with healthcare regulations.
 - Implement Azure Compliance Manager to manage compliance documentation and activities.

- Monitoring and Auditing:
 - Use Azure Monitor and Azure Log Analytics for real-time monitoring of operations and health.
 - Configure alerts for anomaly detection and set up audit logs for access and changes to patient data.

6. Infrastructure and Network Layer

- Azure Kubernetes Service (AKS):
 - Deploy microservices such as user authentication and patient data processing within containers for scalability and management.
- Networking:
 - Utilize Azure ExpressRoute for a private connection to Azure services.
 - Apply Azure Application Gateway for load balancing and SSL termination.

7. Disaster Recovery and Backup

- Use Azure Site Recovery for business continuity.
- Schedule regular backups using Azure Backup service.

8. Data Processing and Business Logic

- Use Azure Logic Apps for building automated workflows to process patient data, such as test results and treatment histories.
- Apply Azure Cognitive Services where machine learning is needed, for instance in image recognition tasks for medical imaging.

9. User Interface

- Develop web and mobile applications using Azure Web Apps and Mobile Apps.
- Ensure all applications are compliant with healthcare standards for patient data handling.

10. Development and Operations

- Implement Azure DevOps for continuous integration and deployment (CI/CD) pipelines to streamline updates and maintain the system.
- Use Azure Monitor and Application Insights for monitoring the application's performance and health.

Conceptual Design

1. First is the login page

A Web Page

https://www.samplepr.ca

Electronic Patient Record System



Username

Password

Submit

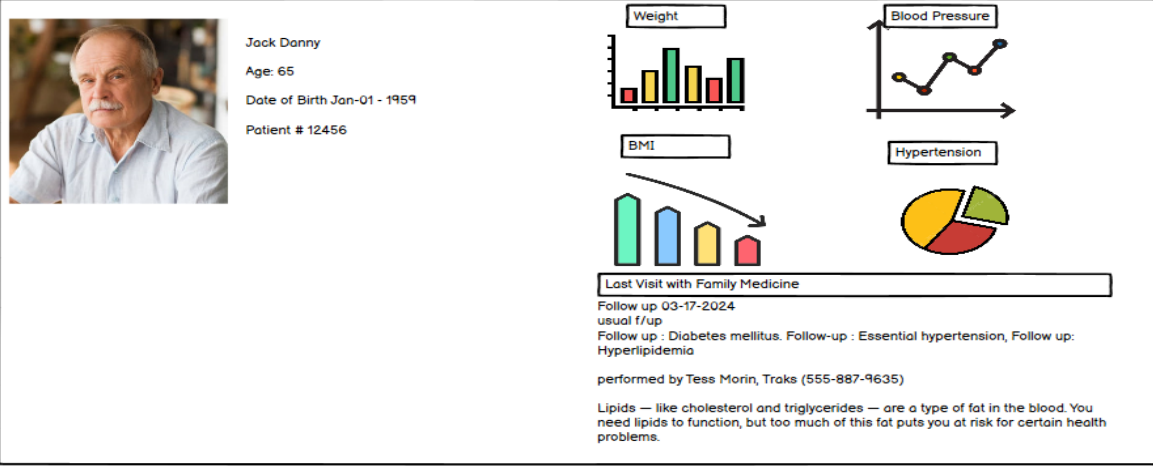
2. The patients profile page

A Web Page

https://www.samplepr.ca

Patient Report Imaging results History Search

Allergens, Problems, Meds, Vaccines, Vitals, Results, History, Quality



Jack Danny

Age: 65

Date of Birth Jan-01 - 1959

Patient # 12456

Weight

BMI

Blood Pressure

Hypertension

Last Visit with Family Medicine

Follow up 03-17-2024

usual f/up

Follow up : Diabetes mellitus. Follow-up : Essential hypertension, Follow up: Hyperlipidemia

performed by Tess Morin, Traks (555-887-9635)

Lipids — like cholesterol and triglycerides — are a type of fat in the blood. You need lipids to function, but too much of this fat puts you at risk for certain health problems.

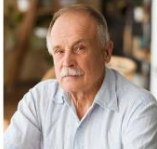
5. Patients treatment history page

A Web Page

https://www.samplepr.ca

Patient Report Imaging results History Search

Allergens
Problems
Meds
Vaccines
Vitals
Results
History
Quality



Jack Danny
Age: 65
Date of Birth Jan-01 - 1959
Patient # 12456

Last Visit report
Follow-up on 02/03/2024
Sample Written History and Physical Examination
History and Physical Examination Comments
Patient Name: Rogers, Pamela
Date: 6/2/04
Referral Source: Emergency Department
Data Source: Patient
Chief Complaint & ID: Ms. Rogers is a 56 y/o WF Define the reason for the patient's visit as who has been having chest pains for the last week, specifically as possible.

History of Present Illness
This is the first admission for this 56 year old woman, convey the acute or chronic nature of the problem and who states she was in her usual state of good health until establish a chronology one week prior to admission. At that time she noticed the abrupt onset (over a few seconds to a minute) of chest pain onset which she describes as dull and aching in character. The character pain began in the left para-sternal area and radiated up to location her neck. The first episode of pain one week ago occurred radiation when she was working in her garden in the middle of the circumstances, exacerbating factors. She states she had been working for approximately 45 minutes and began to feel tired before the onset of the pain.

[Treatment History link](#)

Health History Form

The information reported below will assist in our history taking only. It is not to ask any questions about the information being requested. Please note that all information provided below will be kept confidential unless allowed or required by law. Your written permission will be requested before we use information for any other purpose.

Name: _____ Phone #: _____
Address: _____ Date of Birth: _____
Occupation: _____
Have you received marriage therapy before? ☐ Yes ☐ No
Did a health care professional ever give you marriage therapy? ☐ Yes ☐ No
If yes, please provide their name and address: _____

Do you indicate conditions you are experiencing or have experienced?

Cardiovascular	Respiratory	Head/Neck
High blood pressure Low blood pressure Chronic congestive heart failure Heart attack Atrial fibrillation / irregular pulse Stroke/TIA Pneumonia or similar illness Heart disease	Asthma COPD Cystic Fibrosis Emphysema Chronic bronchitis Other lung conditions	History of head/neck injury or surgery Swallowing problems Voice loss Ear problems Hearing loss
Is there a family history of any of the above? ☐ Yes ☐ No	Other Conditions List all conditions, when? diabetes, cancer, arthritis, hypertension, etc. Is there a family history of any of the above? ☐ Yes ☐ No	Women pregnancy, date, complications, conditions, other Obstetrical history of your general health?

Current Medications
List all medications, when?
condition it treats
Do you have any medical conditions (e.g. diabetes, hypertension, osteoporosis, mental illness)? ☐ Yes ☐ No
If yes, for what?

Do you have any internal pills, shots, infused fluids or special equipment? ☐ Yes ☐ No
If yes, for what?

What is the reason you are seeking marriage therapy?
Please indicate the location of any stress or past difficulties.

Support - date: _____
name: _____
address: _____

Storing patient's medical records in Azure Portal

1. Creating containers for securely storing patient records.

portal.azure.com/#home

Microsoft Azure

Search resources, services, and docs (G+/I)

moht1076@algonquin...
ALGONQUIN COLLEGE, ALGONQUIN

Azure services

Create a resource
Storage accounts
Microsoft Entra ID
Quickstart Center
Virtual machines
App Services
SQL databases
Azure Cosmos DB
Kubernetes services
More services

Resources

Recent Favorite

Name	Type	Last Viewed
gldcloudstorage	Storage account	a minute ago
Group1_group	Resource group	18 hours ago

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Subscriptions
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Tools

Microsoft Learn
Learn Azure with free online training from Microsoft

Azure Monitor
Monitor your apps and infrastructure

Microsoft Defender for Cloud
Secure your apps and infrastructure

Cost Management
Analyze and optimize your cloud spend for free

Useful links

Technical Documentation
Azure Migration Tools

Azure Services
Find an Azure expert

Recent Azure Updates
Quickstart Center

Azure mobile app

Download on the App Store
GET IT ON Google Play

2. Configuring the storage containers based on HIPAA regulations.

The screenshot displays the 'Overview' page for the storage account 'g6cloudstorage' in the Microsoft Azure portal. The left-hand navigation pane is expanded to 'Data storage', where 'Containers' is highlighted with a red rectangle. The main content area is divided into several sections:

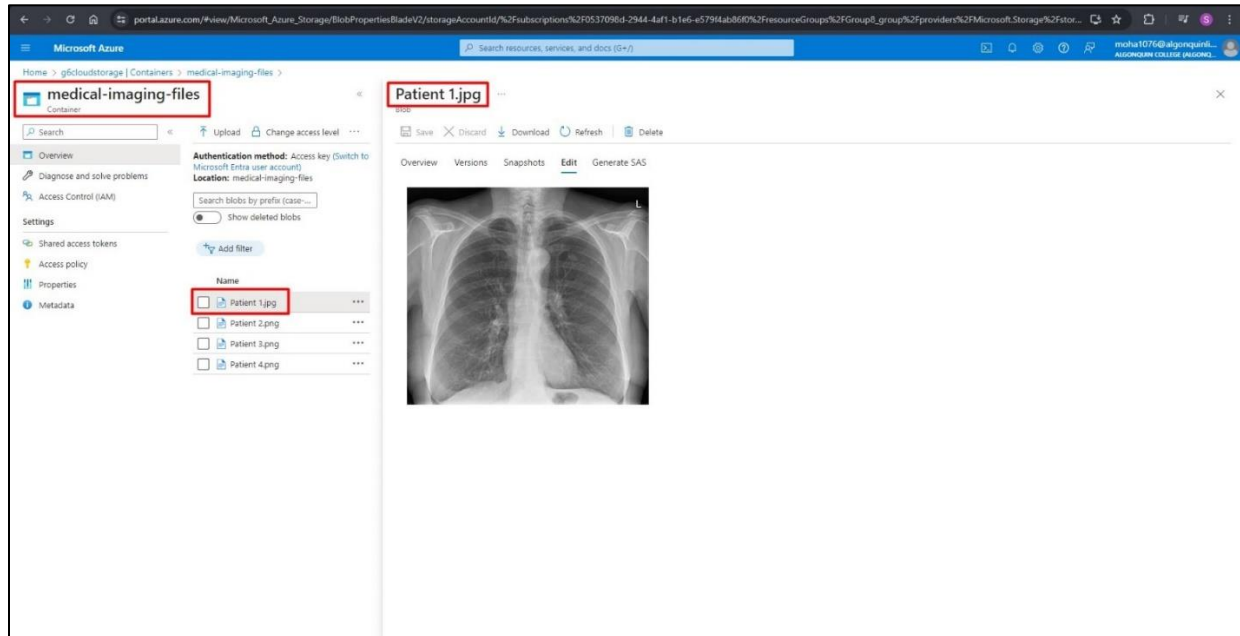
- Essentials:** Provides key information about the storage account, including its resource group ('Group8_group'), location ('eastus'), subscription ('Azure for Students'), subscription ID, disk state ('Available'), and tags.
- Properties:** A tab that lists various capabilities and their status. Under the 'Blob service' section, it shows settings like 'Hierarchical namespace' (Disabled), 'Default access tier' (Hot), 'Blob anonymous access' (Disabled), 'Blob soft delete' (Enabled (7 days)), 'Container soft delete' (Enabled (7 days)), 'Versioning' (Disabled), 'Change feed' (Disabled), 'NFS v3' (Disabled), 'Allow cross-tenant replication' (Disabled), and 'Storage tasks assignments' (None). Under the 'File service' section, it shows 'Large file share' (Disabled), 'Identity-based access' (Not configured), 'Default share-level permissions' (Disabled), and 'Soft delete' (Enabled (7 days)).
- Security:** A section detailing security settings, including 'Require secure transfer for REST API operations' (Enabled), 'Storage account key access' (Enabled), 'Minimum TLS version' (Version 1.2), and 'Infrastructure encryption' (Disabled).
- Networking:** A section detailing networking settings, including 'Allow access from' (All networks), 'Number of private endpoint connections' (0), 'Network routing' (Microsoft network routing), 'Access for trusted Microsoft services' (Yes), and 'Endpoint type' (Standard).

3. Creating containers for storing different files (imaging file, test results, history files)

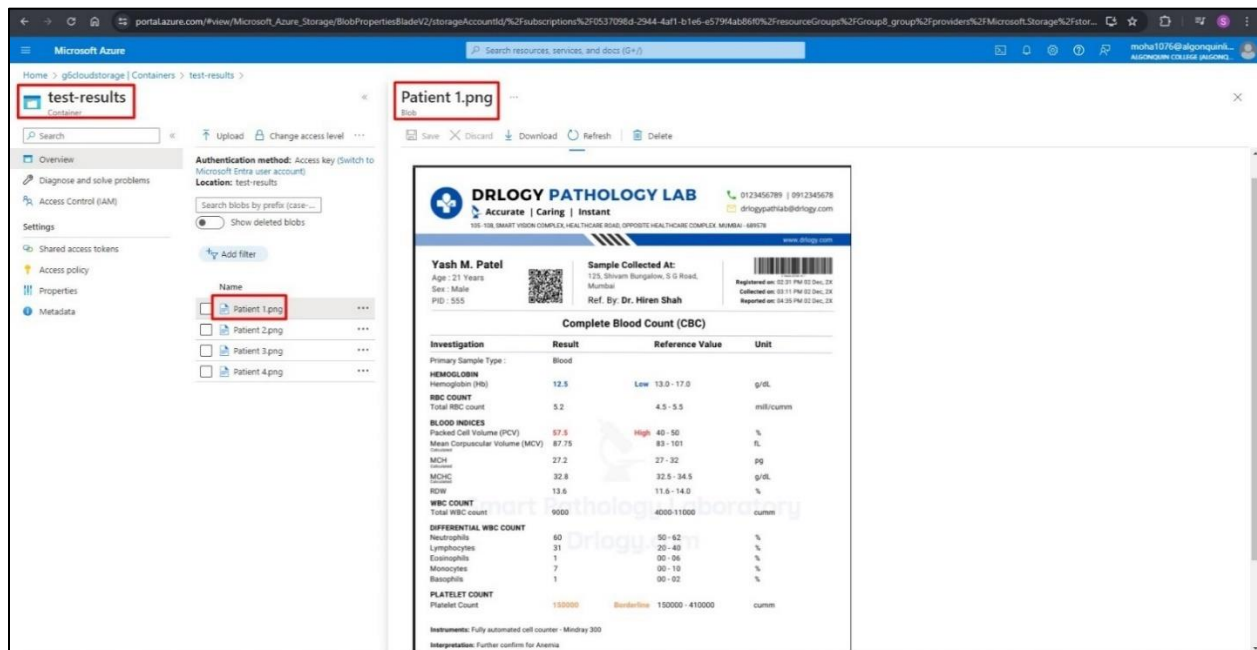
The screenshot displays the 'Containers' page for the storage account 'g6cloudstorage' in the Microsoft Azure portal. The left-hand navigation pane is expanded to 'Data storage', where 'Containers' is highlighted. The main content area shows a table of containers with the following columns: Name, Last modified, Anonymous access level, and Lease state. The table lists four containers: 'blobs', 'medical-imaging-files', 'test-results', and 'treatment-history'. The 'test-results' and 'treatment-history' containers are highlighted in yellow. The 'test-results' container is also highlighted with a red rectangle. The 'test-results' container has a 'Last modified' date of 4/14/2024, 4:24:30 PM, an 'Anonymous access level' of 'Private', and a 'Lease state' of 'Available'.

Name	Last modified	Anonymous access level	Lease state
☐ blobs	4/14/2024, 10:30:44 AM	Private	Available
☐ medical-imaging-files	4/14/2024, 4:25:12 PM	Private	Available
☐ test-results	4/14/2024, 4:24:30 PM	Private	Available
☐ treatment-history	4/14/2024, 4:25:36 PM	Private	Available

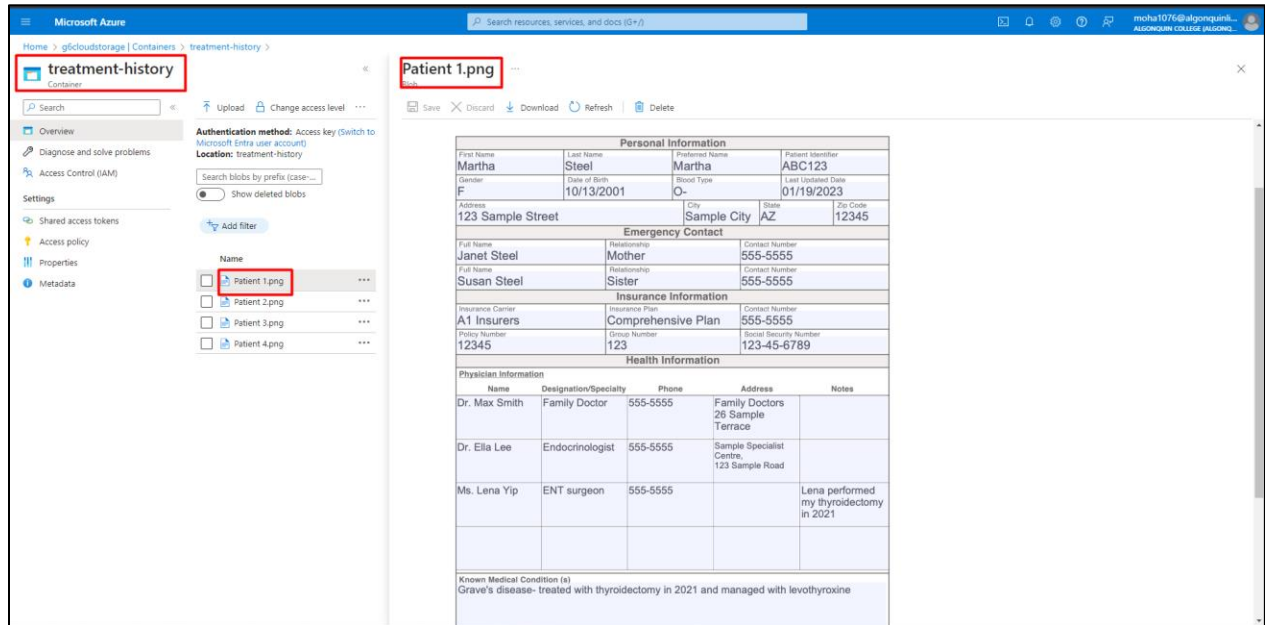
4. The secured storage of patient imaging file in the containers.



5. The secured storage of patient test result file in the containers.



6. The secured storage of patient treatment history file in the containers.



The screenshot shows the Microsoft Azure portal interface. On the left, the 'treatment-history' container is selected, and a list of files is shown, with 'Patient 1.png' highlighted. The main area displays the content of 'Patient 1.png', which is a patient treatment history form.

Personal Information			
First Name	Last Name	Full Name	Patient Identifier
Martha	Steel	Martha Steel	ABC123
Gender	Date of Birth	Blood Type	Last Updated Date
F	10/13/2001	O+	01/19/2023
Address		City	State
123 Sample Street		Sample City	AZ
Zip Code		12345	
Emergency Contact			
Full Name	Relationship	Contact Number	
Janet Steel	Mother	555-5555	
Susan Steel	Sister	555-5555	
Insurance Information			
Insurance Carrier	Insurance Plan	Contact Number	
A1 Insurers	Comprehensive Plan	555-5555	
Policy Number	Group Number	Social Security Number	
12345	123	123-45-6789	
Health Information			
Physician Information		Notes	
Name	Designation/Specialty	Phone	Address
Dr. Max Smith	Family Doctor	555-5555	Family Doctors 26 Sample Terrace
Dr. Ella Lee	Endocrinologist	555-5555	Sample Specialist Centre, 123 Sample Road
Ms. Lena Yip	ENT surgeon	555-5555	Lena performed my thyroidectomy in 2021
Known Medical Condition (s) Grave's disease- treated with thyroidectomy in 2021 and managed with levothyroxine			

Technical Specifications

Application Servers:

- Azure App Services: For hosting applications. Automatically managed, scaled, and patched by Azure.
- Azure Functions: For executing code in response to triggers without provisioning or managing servers.

File Servers:

- Azure Files: Fully managed file shares that use the SMB protocol.
- Azure Blob Storage: Optimized for storing massive amounts of unstructured data.

Web Servers:

- Azure App Services: Again, for hosting web applications with auto-scaling and load balancing features.

Database Servers:

- Azure SQL Database: A fully managed relational database with built-in intelligence and security.
- Azure Database for PostgreSQL/MySQL: Managed database services for PostgreSQL and MySQL.

Server Requests

- Azure API Management: Manage APIs at scale with security, usage policies, and analytics.
- Azure Front Door or Azure Application Gateway: For secure, scalable, and high-performance global entry points for web applications.

Access Requirements

- Azure Active Directory: For secure identity services with multi-factor authentication and conditional access.
- Azure Role-Based Access Control (RBAC): For fine-grained access management to Azure resources.

Databases

- Azure SQL Database: Provides advanced security features that comply with HIPAA regulations, including Transparent Data Encryption (TDE), Always Encrypted, dynamic data masking, and column-level security.
- Azure Cosmos DB: For globally distributed, multi-model database services.

Storage and Compliance

- Azure Storage: Offers encryption for data at rest and in transit, along with access control mechanisms.
- Azure Defender for Storage: Provides advanced threat protection for detecting and responding to potential threats.

Additional Specifications for Compliance

- Azure Policy and Azure Blueprints: To enforce organizational standards and assess compliance at scale.
- Azure Compliance Manager: Helps manage compliance activities, simplifying compliance processes.

Network

- Azure Virtual Network: Creates isolated networks within Azure and connects with on-premises data centers.
- Network Security Groups (NSGs): For filtering network traffic to and from Azure resources.
- Azure VPN Gateway or ExpressRoute: For establishing secure connections between Azure and on-premises networks.

Monitoring and Audit

- Azure Monitor: For full-stack monitoring, advanced analytics, and intelligent insights.
- Azure Audit Logs: To ensure all access and operations are recorded for compliance.

By using the above Azure services, a highly secure, scalable, and compliant Patient Records System is setup. Each service in Azure comes with its own set of configurations that can be tuned to meet the technical and compliance requirements of the healthcare industry.

Resource Requirements

Staff Resources:

- Azure administrators' expertise in cloud computing and healthcare IT.
- Developers proficient in Azure services and application development.
- Security specialists well-versed in HIPAA compliance prerequisites and Azure security functionalities.
- Project managers tasked with supervising the execution and coordination of diverse tasks.

Staff Dependencies:

- Mentors or professors possessing proficient understanding of Azure services and healthcare compliance to offer guidance and assistance.
- Collaboration with healthcare domain experts to ensure synchronization with industry regulations and best practices.
- Access to online communities or forums tailored for Azure professionals and healthcare IT specialists for the purpose of sharing knowledge and resolving issues.

Technical Dependencies:

- Access to Azure subscription resources for deploying and managing the requisite services.
- Provision of development environments and tools suitable for constructing and testing applications, inclusive of Azure App Services and Azure Functions.
- Ensured internet connectivity for accessing Azure services and documentation reliably.
- Proficiency in Azure services and configurations delineated in the technical design document, encompassing Azure Storage, Azure SQL Database, Azure Active Directory, and Azure Security attributes.

Other Dependencies:

- Dedication of time by team members for studying and executing the technical design.
- Allocation of financial resources to procure Azure subscriptions, supplementary services, and any necessary third-party tools.

- Support from the healthcare organization's leadership for determining priorities and distributing resources for the project.
- Availability of testing environments and datasets essential for verifying the functionality and adherence to compliance standards of the Patient Records System.
- Accessibility of training resources and certification programs to guarantee that team members are adequately trained and certified in Azure technologies and healthcare compliance.

Assumptions

1. **Access to Azure Services:** The project presupposes that the team can readily access the essential Azure services outlined in the technical design document, including Azure Storage, Azure SQL Database, Azure App Services, and others.
2. **Technical Proficiency:** It is presumed that the project team possesses the requisite technical expertise necessary to proficiently implement and configure the Azure services as specified in the provided guidelines.
3. **Regulatory Understanding:** The assumption is made that the project team is knowledgeable about healthcare regulations such as HIPAA and can implement Azure services in alignment with these regulatory requirements.
4. **Resource Availability:** It is expected that the project team has access to the necessary resources, encompassing human resources, budgetary allocations, and time, to effectively implement and manage the Azure services.
5. **Security Consciousness:** The project assumes that the team recognizes the critical importance of security in healthcare systems and will undertake the implementation of appropriate security measures to safeguard patient data.
6. **Ongoing Monitoring and Maintenance:** It is anticipated that the project team will engage in continuous monitoring of the Azure services and undertake requisite maintenance activities to uphold the system's reliability, security, and compliance standards.
7. **Integration with Existing Infrastructure:** The project presupposes seamless integration with pre-existing systems within the healthcare organization, including user authentication mechanisms and patient data management systems.
8. **Training and Documentation:** The project team expects that sufficient training and comprehensive documentation will be made available to pertinent stakeholders to ensure the successful implementation and operation of the Azure-based patient records system.

Issues & Concerns

When setting up Azure Storage Account for a healthcare organization, there are several important issues and concerns that need to be addressed to ensure data security, compliance, and optimal functionality. Here are some key considerations:

Data Security:

Ensuring the confidentiality, integrity, and availability of healthcare data stored in Azure Storage is paramount. To achieve this, the project will implement robust security measures, including encryption at rest and in transit, to safeguard sensitive patient information from unauthorized access. Access controls and authentication mechanisms will be configured to restrict access to authorized personnel only, ensuring that patient data remains accessible only to those with proper authorization. Additionally, the project will establish procedures for regularly monitoring and auditing access logs to detect any suspicious activities or unauthorized access attempts promptly. These comprehensive security measures aim to uphold the highest standards of data protection and compliance with healthcare regulations like HIPAA.

HIPAA Compliance:

Healthcare organizations face stringent regulatory requirements, notably those stipulated in the Health Insurance Portability and Accountability Act (HIPAA). Central to this project is ensuring compliance with HIPAA regulations concerning data security, privacy, and breach notification within the Azure Storage setup. This involves the meticulous implementation of administrative, physical, and technical safeguards to safeguard electronic protected health information (ePHI).

Data Governance and Lifecycle Management:

The project encompasses the establishment of robust policies and procedures for data governance, encompassing critical aspects such as data classification, retention, and deletion. To achieve this, lifecycle management strategies will be implemented to automate the archiving or deletion of data in accordance with predefined retention policies and compliance requirements. This proactive approach ensures that data is managed efficiently throughout its lifecycle, mitigating risks associated with unauthorized access or unnecessary data retention. Furthermore, stringent access controls will be enforced to restrict modification or deletion permissions to authorized individuals exclusively, thereby bolstering the security and integrity of the stored healthcare data.

Performance and Scalability:

The project involves meticulously designing the Azure Storage architecture to align with the performance and scalability needs of the healthcare organization. This entails optimizing data access patterns and leveraging advanced features such as Azure Blob Storage's tiered storage (hot, cool, archive), tailored to data access frequency and cost considerations. By employing tiered storage solutions, the project aims to efficiently manage data storage costs while ensuring that frequently accessed data remains readily available without compromising performance. Additionally, continuous monitoring of storage performance will be conducted, enabling timely adjustments and resource scaling to accommodate the organization's evolving data volumes and user demands.

Disaster Recovery and Business Continuity:

The project includes the implementation of robust backup and disaster recovery strategies to fortify against data loss and uphold business continuity in the face of outages or unforeseen disasters. Leveraging Azure's built-in redundancy options like geo-replication and zone-redundant storage, data will be replicated across multiple Azure regions, ensuring high availability and resilience to regional failures. Regular testing of disaster recovery procedures will be conducted to validate data recovery capabilities and minimize potential downtime.

Cost Management:

The project encompasses vigilant monitoring and optimization of storage costs to maximize the cost-effectiveness of Azure resources. This involves leveraging Azure Cost Management tools to analyze spending patterns, pinpoint cost-saving opportunities, and fine-tune resource utilization for optimal efficiency. Furthermore, the implementation of cost controls, including budget alerts and resource tagging, will be paramount to preempt unexpected expenses and manage cloud costs effectively.

Projected Costs

The cost of establishing Azure Storage for a healthcare company is determined by considering several variables, including data transmission, operations, redundancy options, and storage type. The possible expenses are broken down as follows:

Azure Storage Reserved Capacity:

Azure Storage Reserved Capacity	Hot	Cold	Archive
100 TB / month	\$2,097	\$1,140	\$123
1 PB / month	\$20,424	\$11,101	\$1,198

Operation and Data Transfer:

Operations and data transfer	Hot	Cold	Archive
Write operations (per 10,000 operations)	\$0.0883	\$0.376	\$0.1765
Read operations (per 10,000 operations)	\$0.0025	\$0.1765	\$0.0821
Data Retrieval	Free	\$0.0408	\$0.0272
Data Transfer (per GB)	\$0.0272	\$0.0272	\$0.0272

Document Tracking

Version 0.1: Security and Data Recovery, the fundamentals:

- As emphasized in the Project Overview and the Technical Design Document, Implementing RDD Recovery method & Zero Trust Framework was the focus at this stage of the project. Implemented following thorough research and discussions regarding the most modern Data-Recovery and Security models in the market. Consequently, enhancing data recovery strategies, and establishing a robust security framework.

Version 0.11: Patient Health Records System using Azure, and compliance with HIPAA:

- Following market assessments and meetings with stakeholders this iteration addressed recognized technical encounters and marked the transition towards a comprehensive patient record system on Azure platform paired by the compliance with HIPAA.

Version 1.0: Simple Document Archive and Secure File Exchange:

- The complete version represents the advanced solution of the Simple-File-Archive and Secure-File-Exchange platform, as outlined in the Technical Design Document. This phase indicated reaching the project's objective by establishing a modernized and tailored environment suitable for health-care secure file archive, recovery, and exchanged. Integrated base on stakeholder's feedback, market study, and project management approval.

Conclusion

The Electronic Patient Records System for Healthcare project on Microsoft Azure represents a significant advancement in managing patient data securely and efficiently. By leveraging Azure's robust cloud infrastructure, the project meets critical HIPAA regulations, ensuring the confidentiality, integrity, and security of patient records. This system not only enhances data accessibility and reliability for healthcare providers but also offers scalable solutions to handle growing data volumes.

The successful implementation of security measures, such as encryption and access controls, alongside comprehensive compliance, and monitoring frameworks, ensures that patient data is protected against unauthorized access and breaches. Overall, this project stands out as a model for future healthcare IT initiatives, aiming to improve patient care through advanced technology while adhering to strict regulatory standards.

Appendix A

Technical Runbook Document for Patient Records System for Healthcare on MS Azure

Introduction

Technical Runbook Document for Patient Records System for Healthcare on Microsoft Azure with HIPAA Regulations

Purpose

This runbook provides comprehensive guidance for establishing, operating, and maintaining a HIPAA-compliant Patient Records System using Microsoft Azure. It is designed to ensure seamless deployment and operation of the system, detailing procedures for routine operations, troubleshooting common issues, and implementing backup and recovery strategies.

Scope

This document outlines the processes for setting up the Azure environment, operational procedures including patient record handling, access management in compliance with HIPAA regulations, troubleshooting common issues, and maintaining data integrity through robust backup and recovery processes.

Environment Setup

Azure Subscription and Resource Group:

- **Azure Subscription:** Verify that you have an active Azure subscription. Create or select an existing resource group specifically for the Patient Records System to simplify resource management.

Azure Services Configuration:

- **Azure App Service:** Deploy an Azure App Service for hosting the patient records application.
- **Azure SQL Database:** Set up an Azure SQL Database for storing patient records with encryption and auditing features enabled to comply with HIPAA requirements.

Network and Security Configuration:

- **Virtual Network:** Configure Azure Virtual Network to ensure isolated and secure network environment.
- **Azure Firewall:** Implement Azure Firewall to protect the network resources.
- **Encryption and Security:** Enable encryption in transit and at rest using Azure's built-in capabilities to meet HIPAA standards.

Operational Procedures

Patient Record Management:

- **Uploading Records:**
 - **Manual Upload:** Utilize the patient records application interface to manually enter or upload patient records.
 - **Automated Upload:** Develop or implement existing APIs for secure and automated data entry from integrated healthcare systems.

User and Access Management:

- **Azure Active Directory (AD):** Use Azure AD for managing healthcare staff identities and credentials.
- **Role-based Access Control (RBAC):** Implement RBAC to define and assign roles ensuring that staff access is restricted to the minimum necessary information as per HIPAA.

Record Retrieval and Maintenance:

- **Retrieval Interface:** Access patient records securely through the patient records application with necessary authentication.
- **Data Integrity and Updates:** Implement data validation checks within the application to ensure the accuracy and completeness of the records.

Troubleshooting Guide

Common Issues:

1. **Issue: Patient records not updating:**
 - **Resolution:** Verify that the application services are running correctly. Check database connectivity and permissions.
2. **Issue: Access denied errors:**
 - **Resolution:** Ensure that roles and permissions are appropriately assigned in Azure AD and RBAC settings are correctly implemented.
3. **Issue: Data synchronization issues:**
 - **Resolution:** Check the API and network configurations. Ensure that synchronization services are operational.

Backup and Recovery

Data Backup:

- **Regular Backups:** Configure Azure Backup to automatically perform regular backups of the Azure SQL Database and any other critical data stores.
- **Geo-Redundancy:** Enable geo-redundant storage to ensure availability and durability of backups across multiple geographic locations.

Data Recovery:

- **Recovery Services:** Utilize Azure Site Recovery for emergency recovery scenarios to minimize downtime and data loss.
- **Test Restores:** Periodically perform test restores of backups to ensure that they are effective and that recovery procedures are well-understood and documented.

Conclusion

This runbook acts as an essential guide for healthcare IT administrators and professionals responsible for managing a HIPAA-compliant Patient Records System on Microsoft Azure. It provides the necessary steps and procedures to ensure the confidentiality, integrity, and availability of patient records while complying with regulatory standards.

Appendix B

Technical As-Built Document for Patient Records System for Healthcare on MS Azure

Project Overview

System Description

The Patient Records System (PRS) is designed to store, manage, and access patient health information in compliance with the Health Insurance Portability and Accountability Act (HIPAA). This system is hosted on Microsoft Azure, leveraging various Azure services to ensure security, reliability, and compliance with healthcare regulations.

Objectives

- **Secure Storage:** Ensure all patient data is stored securely using encryption both at rest and in transit.
- **Accessibility:** Provide healthcare professionals with reliable and secure access to patient records.
- **Compliance:** Meet all HIPAA requirements for privacy and security.

Scope

This document covers the technical specifications, configurations, and components used in the PRS deployed on Microsoft Azure.

Technical Specifications

Azure Environment Setup

Subscription Details

- **Subscription ID:** [Redacted]
- **Resource Group:** PRS-Resources

Core Services

- **Azure Active Directory (AAD):** Used for identity management and access control.
- **Azure App Service:** Hosts the web application for patient record management.
- **Azure SQL Database:** Stores patient records with Transparent Data Encryption (TDE) enabled.

Network and Security Configuration

Virtual Network

- **Name:** PRS-VNet
- **Subnets:**
 - Frontend (App Service)
 - Backend (SQL Database)
 - Management (Administrative Access)
- **Network Security Groups (NSGs):** Configured to restrict traffic based on least privilege access.

Firewall and Security

- **Azure Firewall:** Configured to monitor and control inbound and outbound traffic.
- **Application Gateway:** Provides application-level routing and load balancing.

Compliance and Security Measures

Encryption and Data Protection

- **Encryption at Rest:** Implemented using Azure Disk Encryption and TDE for SQL Database.
- **Encryption in Transit:** Enforced TLS 1.2 for all connections.

Identity and Access Management

- **Azure AD Configurations:**
 - Multi-Factor Authentication (MFA) enabled for all users.
 - Role-Based Access Control (RBAC) implemented to enforce least privilege.

Auditing and Monitoring

- **Azure Monitor:** Used to track performance and health of services.
- **Azure Security Center:** Provides unified security management and advanced threat protection.

Application Architecture

Components

- **Web Application:** ASP.NET Core application hosted on Azure App Service.
- **Database:** Azure SQL Database with row-level security and dynamic data masking.
- **APIs:** RESTful services for secure data access and integration with other systems.

Backup and Disaster Recovery

Backup Configurations

- **Azure Backup:** Scheduled daily backups for Azure SQL Database and Web Apps.
- **Storage Replication:** Geo-redundant storage enabled for backups.

Disaster Recovery Strategy

- **Azure Site Recovery:** Configured for automated disaster recovery.
- **Recovery Plan:** Documented and tested annually to ensure effective restoration capabilities.

Conclusion

This as-built document provides a detailed overview of the Patient Records System deployed on Microsoft Azure, designed to meet the stringent requirements of HIPAA. It serves as a comprehensive reference for system specifications, security measures, and compliance strategies, ensuring that the system remains secure, compliant, and operationally effective.

Appendix C

Management of Risks

1. Risk Identification:
 - a. To Identify the potential risks related to data security, compliance, performance, scalability, disaster recovery, and cost management. Also Consider factors such as unauthorized access, data breaches, non-compliance with HIPAA regulations, storage performance bottlenecks, unexpected downtime, and budget overruns.
2. Risk Assessment:
 - a. By evaluating the impact and likelihood caused by each identified risk on the Azure Storage setup and the overall patient records system.
 - b. Prioritize risks based on their severity and potential consequences for patient data security, regulatory compliance, system availability, and financial implications.
3. Risk Mitigation:
 - a. Develop risk mitigation strategies to address identified risks effectively.
 - b. Implement security measures such as encryption, access controls, and monitoring to protect patient data from unauthorized access and breaches.
 - c. Ensure compliance with HIPAA regulations by following best practices for data security, privacy, and breach notification.
 - d. Optimize storage performance and scalability by configuring Azure Storage tiers, monitoring usage patterns, and scaling resources as needed.

- e. Establish robust disaster recovery and business continuity plans to minimize downtime and data loss in case of outages or disasters.
 - f. Monitor and optimize storage costs to prevent budget overruns and ensure cost-effective utilization of Azure resources.
4. Risk Monitoring and Review:
- a. Continuously monitor the Azure Storage setup for any emerging risks or changes in risk factors.
 - b. Regularly review and update risk management strategies and controls to adapt to evolving threats and organizational requirements.
 - c. Conduct periodic risk assessments and audits to verify the effectiveness of risk mitigation measures and ensure ongoing compliance with regulations and best practices.
5. Stakeholder Communication and Engagement:
- a. Maintain open communication channels with stakeholders, including healthcare professionals, IT staff, management, and regulatory authorities.
 - b. Keep stakeholders informed about identified risks, mitigation efforts, and any changes in the risk landscape.
 - c. Encourage collaboration and feedback from stakeholders to enhance risk management practices and address emerging challenges effectively.

Stakeholders

Stakeholder Names	Roles	Stakeholder Influence
Ahmad Touseef	Project Sponsor	High
John Dallas	Sponsor	High
Arsalan Parsaei	Sponsor	Medium
Clients	Sponsor	High
Charles Nixon	IT Support	Medium
Stanley Desmoulins	Product Owner / Sponsor	High
Kazi Tahzid Irteza	Team Lead	Medium
Shaik Shahil Sheriff	Team Member	Medium
Oluwatoni Jason Oderinlo	Team Member	Medium
Mohamed Mahmoud	Team Member	Medium
Tarek Mohammed	Team Member	Medium
MS Azure Support team	Support	High

Communication & Changes

Version	Capstone Project Name	Description
0.1	Implementing RDD Recovery and Zero Trust Security Framework	This version focuses on implementing RDD recovery and a Zero Trust Security Framework for enhanced security.
0.11	Patient Records System for Health with HIPAA and Azure	This version aims to address long-term technical issues by providing a simple document archive and secure file exchange platform for better collaboration.
1	Simple Document Archive and Secure File Exchange	This version involves the development of a patient records system compliant with HIPAA regulations, leveraging Azure services for enhanced security and compliance.

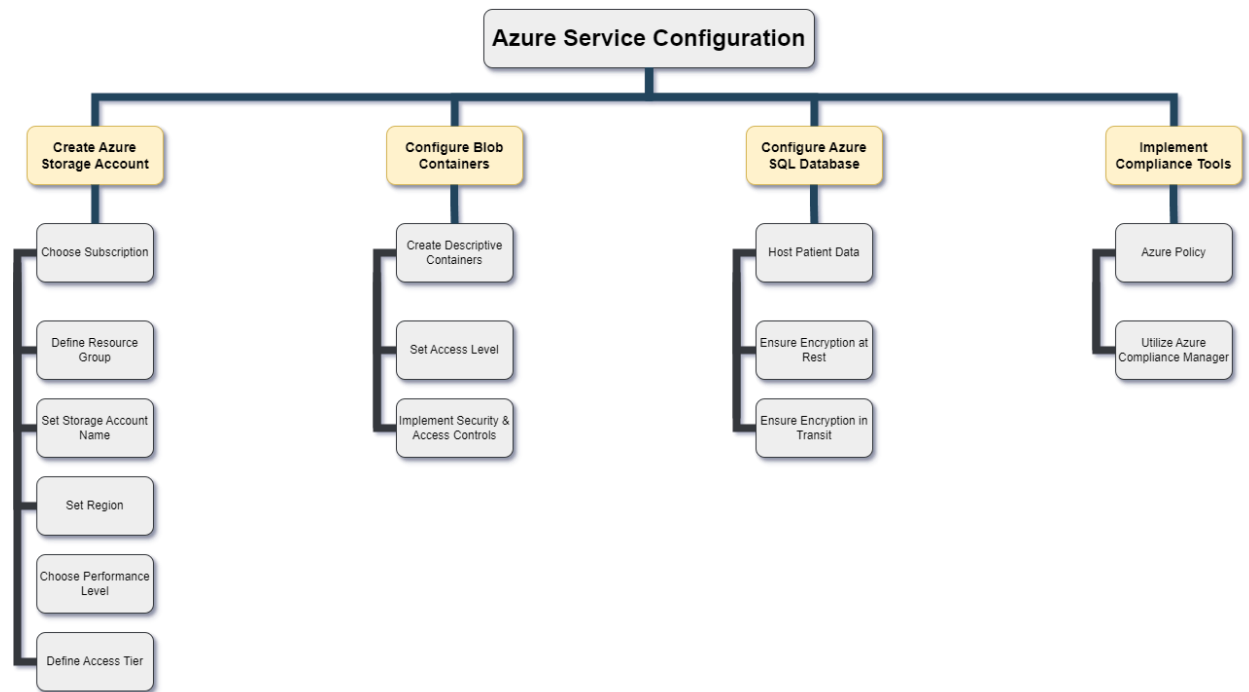
Initial Scope

The initial scope of the project encompasses a comprehensive exploration of Azure management and governance functionalities. This includes identity management, governance through Management Groups and RBAC roles, consistent resource tagging, efficient resource organization, deployment using ARM templates, and resource management through Azure PowerShell and CLI. Additionally, the scope extends to Azure networking aspects such as virtual network creation, VM deployment, IP configuration, security groups, and DNS setup. Subsequent labs cover virtual network peering, hub-and-spoke topology, Azure Load Balancer, Application Gateway, Azure Storage management, and Azure virtual machines. Furthermore, the project encompasses Azure web app management, Docker image deployment, resource group deletion, backup strategies, and environment provisioning with Log Analytics workspace creation.

Initial Planning

The initial planning phase involves each lab be assigned a specific timeframe for completion, considering the complexity of tasks involved and the learning objectives. Resources required, including Azure subscriptions, virtual machines, storage accounts, and networking resources, will be provisioned in advance to ensure smooth execution of the labs. Additionally, lab environments will be set up with necessary tools such as Azure PowerShell, CLI, and ARM templates readily available. Training materials, documentation, and support resources will be prepared to assist participants throughout the labs. A communication plan will be established to keep all stakeholders informed about the project's progress and any updates or changes to the plan. However, due to technical difficulties, we later received approval from the stakeholders to shift our work from Project AZ-104 to Project Electronic Patient Records System for Healthcare.

Work Breakdown Structure



Acceptance Test

Testcase ID	Objective	Steps	Expected Results
FR01	Ensure new patient records can be added by authorized personnel.	1. Log in as an authorized healthcare provider. 2. Navigate to the 'Add New Patient' section. 3. Enter required patient information including name, date of birth, medical history, and contact details. 4. Submit the information.	A new patient record is created without errors. The user receives a confirmation message that the patient record has been added.
FR02	Verify healthcare providers can access and review the treatment history of a patient.	1. Log in as a healthcare provider. 2. Search for a specific patient using the patient ID or name. 3. Select the patient from the search results. 4. Navigate to the 'Treatment History' tab.	The treatment history for the selected patient is displayed. The information includes all past and ongoing treatment records.

FR03	Confirm that patient information can be updated correctly by authorized users.	1. Log in as a healthcare provider. 2. Locate an existing patient record. 3. Select the option to edit patient information. 4. Change several fields (e.g., address, emergency contact). 5. Save the changes.	The patient information is updated successfully. Changes are reflected immediately when the record is viewed again.
FR04	Ensure that patient records can be deleted by authorized personnel with proper logging.	1. Log in as an administrator. 2. Search for the patient record to be deleted. 3. Select the delete option and confirm the action.	The patient record is permanently deleted from the system. An audit trail entry is created logging the deletion.
FR05	Test the secure messaging functionality for communication between healthcare providers.	1. Log in as a healthcare provider. 2. Navigate to the secure messaging module. 3. Compose a message to another provider, attaching relevant patient information. 4. Send the message.	The message is sent successfully without errors. The recipient receives the message and can view the attached patient information securely.
UAT01	Verify login functionality and user-specific accessibility.	1. Log in with valid healthcare provider credentials. 2. Attempt to access provider-specific functionalities.	Successful login. Access to provider-specific functions is verified without error.
UAT02	Ensure patient records can be updated and saved correctly.	1. Log in as administrative staff. 2. Access a patient's record. 3. Update contact details and medical notes. 4. Save changes.	Patient record updates correctly and shows updated details without requiring refresh.
UAT03	Test system backup and data recovery procedures.	1. Perform a routine backup of data. 2. Simulate a data loss scenario. 3. Attempt to restore data from backup.	Backup completes successfully. Data restores accurately and within acceptable time.
UAT04	Confirm secure access and functionality of the medical	1. Log in as a healthcare provider. 2. Access a patient's medical imaging files. 3. View and annotate images.	Medical imaging files are accessible, display correctly, and annotations are saved

	imaging files.		properly.
UAT05	Validate compliance with HIPAA privacy and security rules.	1. Audit user access logs. 2. Review encryption standards for data at rest and in transit. 3. Check adherence to data access protocols.	All system functionalities comply with HIPAA regulations. Access logs and encryption protocols meet the set standards.

Lessons Learned

We've learned invaluable knowledge from our project, which focused on project management, presenting techniques, and teamwork. These teachings go beyond the classroom and directly relate to real-world situations. Above all, we've learned how to differentiate between careful planning and flexibility. Although project management frameworks offer an organized method, it is equally important to be able to change course and adapt plans in the face of unforeseen difficulties. Furthermore, developing one's presenting abilities has highlighted how crucial audience participation, conciseness, and clarity are. Moreover, cultivating efficient team communication has been crucial in managing varied viewpoints and utilizing group proficiency to achieve common goals.

Training Plan

The Training Plan created and needed for the project should encompass the following:

1. **Azure Services and Configuration:** Offer comprehensive training on Azure services pertinent to the project, including Azure Storage, Azure SQL Database, Azure App Services, Azure Functions, and Azure Active Directory. Ensure comprehension of configuration settings, security features, and compliance considerations specific to healthcare regulations like HIPAA.
2. **Data Security and Compliance:** Provide specialized training on data security best practices, encryption methods, access controls, and compliance requirements, with a particular focus on HIPAA regulations and Azure security features such as encryption at rest, Azure Information Protection, and Azure Key Vault.
3. **Disaster Recovery and Business Continuity:** Educate team members on disaster recovery strategies, encompassing Azure Site Recovery, backup and restore procedures, failover planning, and testing methodologies. Stress the significance of maintaining business continuity in healthcare environments and ensuring uninterrupted access to patient records.
4. **Performance Optimization and Scalability:** Deliver training on optimizing Azure Storage performance, monitoring storage usage, effectively leveraging Azure Storage tiers, and

scaling resources based on demand. Cover topics related to performance tuning, load balancing, and resource optimization to ensure efficient operation of the patient records system.

5. **Cost Management:** Educate team members on cost management strategies for Azure services, including monitoring usage, implementing cost controls, optimizing resource utilization, and effectively utilizing Azure Cost Management tools. Highlight the importance of cost-conscious decision-making to prevent budget overruns and ensure cost-effective use of Azure resources.
6. **Security Awareness and Training:** Conduct security awareness training sessions to enlighten team members about common security threats, social engineering attacks, phishing attempts, and best practices for safeguarding sensitive information. Foster a culture of security consciousness and advocate adherence to security policies and procedures.
7. **Compliance Training:** Provide specialized training on healthcare compliance regulations, notably HIPAA, covering topics such as data privacy, confidentiality, breach notification requirements, and penalties for non-compliance. Ensure team members grasp their roles and responsibilities in maintaining compliance with regulatory standards.
8. **Technical Skills Development:** Facilitate hands-on technical training and workshops to enhance team members' proficiency in Azure services, infrastructure management, networking, automation tools, and DevOps practices. Offer opportunities for practical exercises, lab sessions, and real-world simulations to reinforce learning and skill development.
9. **Continuous Learning and Certification:** Encourage team members to pursue relevant Azure certifications, such as the Azure Administrator certification (AZ-104), to validate their skills and expertise. Provide support and resources for exam preparation, including study materials, practice tests, and access to training courses.
10. **Stakeholder Collaboration and Communication:** Foster collaboration and communication among team members, stakeholders, and external partners involved in the project. Encourage regular meetings, status updates, and knowledge sharing sessions to ensure alignment of goals, address concerns, and promote a collaborative working environment.